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FROM FEATURE SPACE TO RELATION SPACE: CLASSIFICATION COMPLEXITY ASSESSMENT BASED ON META-OBJECTS

¹Ignatev N, ²Tuhtabayev K

¹ *National University of Uzbekistan named after Mirzo Ulugbek, Vuzgorodok 4, Tashkent, 100174, Uzbekistan,*
e-mail: n_ignatev@rambler.ru

² *National University of Uzbekistan named after Mirzo Ulugbek, Vuzgorodok 4, Tashkent, 100174, Uzbekistan,*
e-mail: ktuhtabayev@gmail.com

This paper proposes an approach to assessing the complexity of classification problems based on the construction of synthetic features that reflect neighborhood relations between objects in the data space. The synthetic features are constructed using the k -nearest neighbor method under different metrics and are transformed into a unified space by means of feature stability measures. This makes it possible to form meta-objects, that is, representations of training samples suitable for subsequent analysis by machine learning methods. The proposed approach is aimed at identifying the structural complexity of data and can be regarded as a development of the combinatorial approach to analyzing training samples on the basis of relations between objects.

Introduction

The construction of synthetic features is one way of adapting the original data to a representation that is more closely related to the classification problem being solved. In contrast to the selection of already existing features, synthetic feature construction involves creating new characteristics that differ from the original description of objects [1–3]. Modern representation learning theory also considers the quality of the derived feature space as an important factor affecting the results of subsequent analysis.

Nearest neighbor methods use local relations between objects, which determine the structure of a sample on the basis of a distance measure. In this case, the geometry of neighborhood relations depends on the measurement scales of the features, the dimensionality of the space, and the selected metric. Differences between metrics become especially significant in multidimensional spaces.

The variety of forms for representing neighborhood relations based on distance measures, determined through k -nearest neighbors, may be expressed:

- in the form of a binary sequence reflecting the results of applying the majority rule over a system of nested hyperspheres;
- in the form of a sequence of nearest-neighbor categories over a system of nested hyperspheres;
- through the number of objects of the same class among the nearest neighbors.

Synthetic features constructed through k -nearest neighbors actually encode relations between objects rather than their original values. Therefore, they can be interpreted as a transition:

feature space $\longrightarrow$ relation space.

In the works of Yu. I. Zhuravlev's school, the data structure is considered through systems of relations between objects, partitions, and transformations of feature descriptions [4–5]. These studies provide a methodological basis for moving from individual original values to characteristics of relations between objects.

The solution of classification problems is closely related to the concepts of generalization ability, the compactness hypothesis, and the complexity of recognition algorithm models. With respect to model complexity, there are two approaches:

- assessment of model capacity in the VC space, that is, Vapnik–Chervonenkis theory [6], based on the mathematical framework of uniform convergence of event occurrence frequencies over the number of realizable partitions to probabilities;

- assessment of combinatorial complexity taking into account hidden regularities in the data [7]. Combinatorial assessment makes it possible to significantly reduce the size of the training sample and to obtain guaranteed recognition accuracy.

If the VC dimension answers the question “How rich is the class of algorithms?”, then Vorontsov’s combinatorial complexity [7] rather answers the question “How complex is the structure of a particular sample, and what relations between objects must be taken into account for its correct recognition?”

The originality of Vorontsov’s approach lies in the introduction of various characteristics of the combinatorial complexity of a training sample, based on:

- the number of admissible partitions of objects;
- the structure of relations between objects;
- systems of support sets;
- tests and coverings;
- the local geometry of the sample.

For estimate computation algorithms and algebraic methods of recognition, the following question is considered: How complex is this particular training sample for separating the classes? In contrast to VC theory, the object of study here is not only a family of decision functions, but also the configuration of the data itself.

An approach to assessing the complexity of data classification problems using the framework of fuzzy set theory is proposed. Membership functions and feature stability measures are considered as research tools [8]. Feature stability is used to form a unified feature space intended for describing meta-objects.

A meta-object is understood as a representation of a training sample in a unified feature space, to which classical machine learning methods can be applied for analysis.

The main scientific novelty of this work is as follows:

1. A method is proposed for constructing binary sequences to represent synthetic features based on the majority rule.
2. A method is proposed for ordering feature sequences according to a system of nested hyperspheres formed using k -nearest neighbors.
3. An approach is developed for forming a set of meta-objects based on different distance metrics that determine the results of majority voting, followed by a transition to a unified space according to feature stability values.
4. A data complexity assessment for classification problems is proposed based on the analysis of the formed meta-objects.

Mathematical Model

Let

$$E_0 = \{S_1, \dots, S_m\}$$

be a finite labeled sample divided into two non-overlapping classes K_1 and K_2 , with

$$K_1 \cup K_2 = E_0.$$

An object S_i is described by a set of quantitative features

$$X(n) = (x_1, \dots, x_n).$$

Let a finite set of metrics

$$P = \{\rho\}$$

be defined on the feature set $X(n)$.

It is required to:

- develop an algorithm for constructing binary sequences based on a system of nested hyperspheres and the set of k -nearest neighbors for each metric $\rho \in P$, forming a set of synthetic features

$$A(t) = (a_1, \dots, a_t);$$

- implement a procedure for forming meta-objects based on object descriptions in the space of synthetic features $A(t)$;
- compute and analyze data classification complexity measures for different metrics from the set P .

Algorithm for Constructing Bit Sequences Based on the Majority Rule

For an object S belonging to class K_t , where $t \in \{1, 2\}$, the k -nearest neighbors are determined according to the metric ρ . In the case of equal distances, the objects are ordered by increasing indices.

For each value of k from the given range $[k_{\min}; k_{\max}]$, the value $a_k(S)$ is computed, which is equal to the number of objects of class K_t among the k -nearest neighbors of object S . The obtained values form the vector

$$A(S) = (a_{k_{\min}}(S), a_{k_{\min}+1}(S), \dots, a_{k_{\max}}(S)).$$

On the basis of this vector, a bit sequence is constructed according to the majority rule. For each value of k , the bit value is determined by the rule

$$b_k(S) = \begin{cases} 1, & \text{if } a_k(S)/k > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

The sequence

$$(b_{k_{\min}}(S), \dots, b_{k_{\max}}(S))$$

is considered as a synthetic feature of object S .

For the purposes of analysis, it is reasonable to:

1. Construct the distributions of the obtained binary sequences by classes K_1 and K_2 , interpreting each sequence as a binary number with subsequent conversion into decimal representation for the values of k from the given range $[k_{\min}; k_{\max}]$.
2. Identify the binary sequences for which the distributions of K_1 and K_2 in the sample E_0 differ most significantly, and also investigate the influence of the choice of metric from the set P on the nature of these differences.

Complexity Assessment of the Classification Problem

Based on the frequencies of occurrence of synthetic feature values in the descriptions of objects belonging to the classes, feature stability vectors are computed and are further interpreted as meta-objects. An algorithm is proposed for assessing the complexity of a classification problem based on the descriptions of meta-objects.

For each feature a_i , where $i \in \{1, 2, \dots, t\}$, and for each of its values, or gradations, μ , we define:

- $d_{1i}(\mu)$ — the number of objects of class K_1 for which $a_i = \mu$;
- $d_{2i}(\mu)$ — the number of objects of class K_2 with the same value μ ;

- $n(\mu) = d_{1i}(\mu) + d_{2i}(\mu)$;
- $h = |\{\mu : n(\mu) > 0\}|$ — the number of different gradations of the feature;
- different gradations are assigned numbers from the set $\{1, \dots, h\}$.

The stability measure g_i is computed using the following formulas:

$$f_i(\mu) = \frac{d_{1i}(\mu)/|K_1|}{d_{1i}(\mu)/|K_1| + d_{2i}(\mu)/|K_2|}, \quad (1)$$

$$g_i = \frac{1}{m} \sum_{\mu=1}^h \begin{cases} n(\mu) \cdot f_i(\mu), & f_i(\mu) \geq 0.5, \\ n(\mu) \cdot (1 - f_i(\mu)), & f_i(\mu) < 0.5. \end{cases} \quad (2)$$

The meta-object

$$Q = (g_1, \dots, g_t)$$

is considered as a vector of aggregated fuzzy characteristics, where its component g_i reflects the degree of stable separation of the two classes by feature a_i . The complexity assessment of the classification problem is defined by the expression:

$$C(Q) = 1 - \frac{1}{t} \sum_{i=1}^t g_i. \quad (3)$$

The following linguistic interpretation of the complexity assessment value $C(Q)$ can be proposed:

- $C(Q) \in (0; 0.25]$ — the problem is simple for classification;
- $C(Q) \in (0.25; 0.5]$ — the problem is complex for classification.

For the set of meta-objects computed by formulas (1) and (2) for all metrics $\rho \in P$, various analysis methods can be used, including the calculation of pairwise correlation coefficients and visualization by the principal component analysis method. Estimate (3) can be applied in classification and regression problems when justifying the choice of a metric, a data normalization method, and methods for selecting informative features.

Computational Experiment

The Hypertension dataset [9] was used as the experimental dataset. To describe 147 objects, 29 quantitative features were used. Class K_1 (controls) contained 111 objects, while class K_2 (patients/affected cases) contained 36 objects.

Synthetic features were computed for values of k from the set

$$R = \{3, 5, \dots, z\},$$

where

$$z = 2 \min(|K_1|, |K_2|) - 3.$$

To compute the six meta-objects, the Euclidean, Chebyshev, and Canberra metrics were used in the original feature space and in the feature space normalized to the interval $[0; 1]$. The principal component analysis (PCA) method was used to display, or visualize, the meta-objects in a two-dimensional space. The visualization results are presented in Fig. 1.

The classification complexity estimates (3), obtained from the descriptions of six meta-objects using synthetic features, are presented in Table 1.

Table 1. Classification complexity assessment on the Hypertension dataset

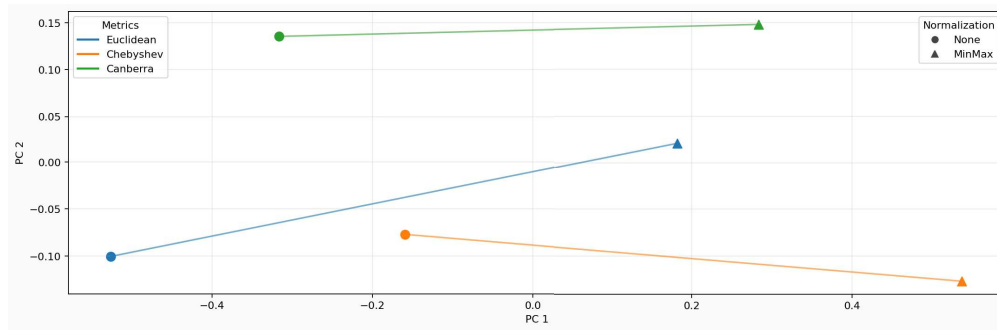


Fig. 1. Visual representation of meta-objects on a plane

Metric	Original space	Normalized space	Difference
Euclidean	0.38326	0.26452	-0.11874
Chebyshev	0.32151	0.20762	-0.11389
Canberra	0.34464	0.25460	-0.09005

The visual representation (Fig. 1) clearly demonstrates differences in the structures of relations between objects both when using different distance measures and when applying data normalization. The transition to the normalized space (see Table 1) leads to a noticeable decrease in the complexity of the classification problem. The magnitude of this effect can be judged by the difference between the complexity estimates obtained in the original and normalized spaces.

Conclusion

This paper proposes an approach to assessing the complexity of classification problems based on the transition from the space of original features to the space of relations between objects, defined through a system of k -nearest neighbors.

Based on synthetic features and stability measures, meta-objects were formed, making it possible to obtain a quantitative assessment of classification complexity and to compare different distance metrics and data preprocessing methods.

The results of the computational experiment showed that the choice of metric and feature normalization have a significant effect on the structure of relations between objects and on the final complexity estimates of the problem.

The proposed approach can be regarded as a development of combinatorial methods for analyzing training samples, aimed at studying the data structure rather than only the properties of classification algorithms.

A promising direction for further research is to study the robustness of the proposed complexity estimate, as well as to analyze its asymptotic properties, including the possible convergence of estimates computed from different samples drawn from the same general population to some fixed characteristic of data complexity.

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